is not to say that predators cannot kill and eat larger animals than themselves; a weasel can easily kill a larger rabbit because it is more aggressive. However, an adult lioness, one of the world's top predators, is not capable of killing a healthy adult African elephant. Likewise, a dragonfly larva on the bottom of a pond may be able to prey on a small tadpole, but it would not be able to eat an adult frog. Animals may be capable of killing much smaller prey, but it is simply not worth the effort. Wolves hunt medium-sized or large mammals such as elk. If those mammals disappear from their environment, they find it hard to catch sufficient numbers of smaller animals such as mice to sustain them; the energy they use finding small prey is greater than the energy they gain by consuming them.

Plants cannot run away or fight back, so different considerations apply to herbivores when it comes to food size. There is a maximum size of seed that a given finch, for example, can fit in its bill, so larger finches have an advantage over smaller species. Similarly, individual species of hummingbirds can drink nectar only from flowers up to a certain size, depending on the length of their bill.